

Unlocking insights, shaping agriculture.

As an ABARES mathematician or statistician, you have one of the most important jobs in Australian agricultural research.

Whether it is designing farm surveys, analysing data to identify sector trends, simulating the impacts of different climate scenarios on farm performance, or evaluating the economic cost of an outbreak of a pest or disease, your contributions carry influence.

Multi-disciplinary innovation

You will contribute to multidisciplinary models that integrate statistical, mathematical, and simulation modelling with spatial mapping, invasive species ecology, social science, and economics.

Our mathematicians and statisticians work on projects like the Drought Early Warning System (DEWS) which uses ABARES microsimulation tool [farmpredict](#) to simulate physical and financial outcomes for Australian farm businesses given climate conditions and commodity prices.

As a part of the ABARES team your work will transcend individual disciplines as you tackle issues like the [economic impacts of foot-and-mouth disease](#).

ABARES mathematicians collaborate in-house with our economists to develop models integrating bio-physical and economic principles. They also collaborate with our statisticians to provide real-world data calibration; our invasive species ecologists on epidemiological modelling; and our social scientists assessing the broader societal impacts of outbreak and control measures.

Together, we create a holistic approach to understand and mitigate the complexities of a multitude of challenges.

It's more than just numbers

ABARES farm survey data is the cornerstone for specialised analysis and outputs that underpin the development of government policy. As a specialist in ABARES you'll help develop robust and reliable synthetic data sets from survey data to support spatial analysis, while also protecting the privacy of farmers.

All our outputs are backed by well-documented research and the mathematical and statistical work is no different. We're always finding new ways to improve, discovering and learning alternative methods all in a day's work.

Your work isn't just a job—it's a legacy

By contributing to accurate time series data collected over more than 50 years, you'll be a part of a rich history of agricultural research and analysis.

Each algorithm you develop and every insight you uncover becomes a building block in a foundation that shapes the future of Australian agriculture.

Development opportunities

We're committed to ensuring our specialists have a well-rounded perspective. Our staff take part in a range of development and training opportunities including:

- Preparing presentations and being a part of the ABARES Outlook conference series
- Participating in stimulating and topical projects
- Engaging with the latest modelling software and coding platforms
- Participating in the ABARES in-house training program that takes you through the ins and outs of ABARES and what we do for the sector.

Research Outputs

- [Farm surveys and analysis](#)
- [Farm Data Portal](#)
- [Micro-simulation models of irrigation farms in the southern Murray Darling Basin](#)
- The [Drought Early Warning System Project](#)

You don't have to be an agriculture specialist to work at ABARES, but you do need to be a mathematics or statistics specialist to apply for this work stream.

Dream in number sequences? Visit the [Department's careers page](#) for current opportunities.

Source: <https://www.agriculture.gov.au/abares/careers/abares-mathematicians-and-statisticians>